Topic 1: Bravais Lattices in 3D

Questions we will answer...

- What are the crystal systems and Bravais lattices in 3D?

- How are the different Bravais lattices distinct from each other?

- How do we denote locations, directions and planes in Bravais lattices?

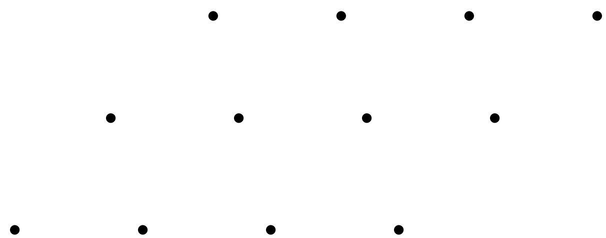
What Is Crystallography?

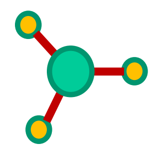
Crystals

Atoms that are bound together ➔ minimize the energy

➔generally achieved by a periodic arrangement

➔ Two elements: **lattice** + **basis**





In 1D…

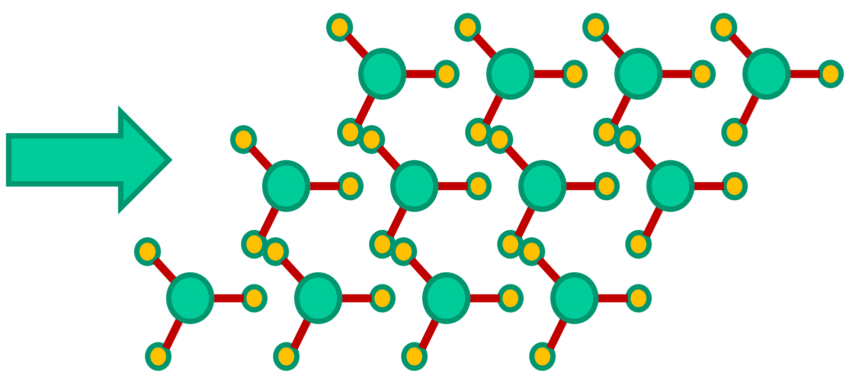
1 Bravais lattice

In 2D…

5 Bravais lattices

In 3D…

14 Bravais lattices



We will revisit 1D & 2D lattices, symmetries, point groups & diffraction in ***Topics 6+***

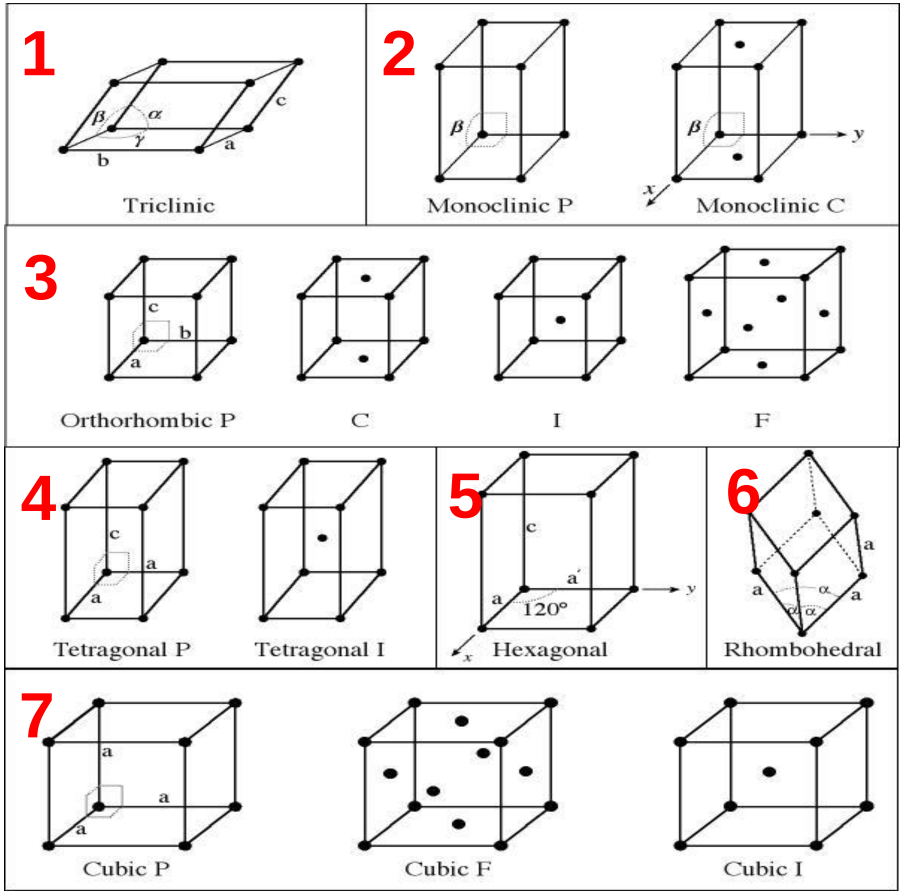
3D Bravais Lattices: Fourteen

7 crystal **systems**

14 crystal **lattices**

P = primitive
C = base-centered
I = body-centered
F = face-centered

Auguste Bravais
1811-1863

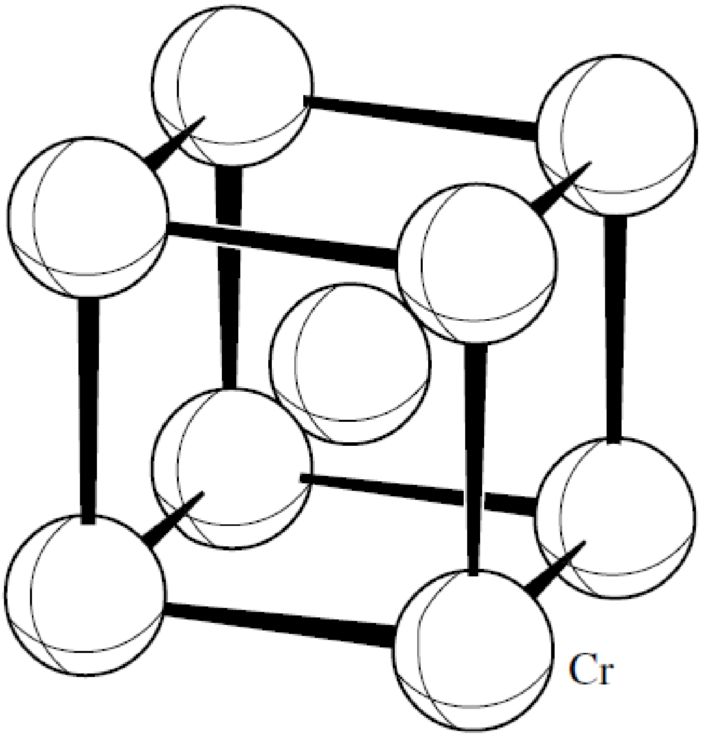


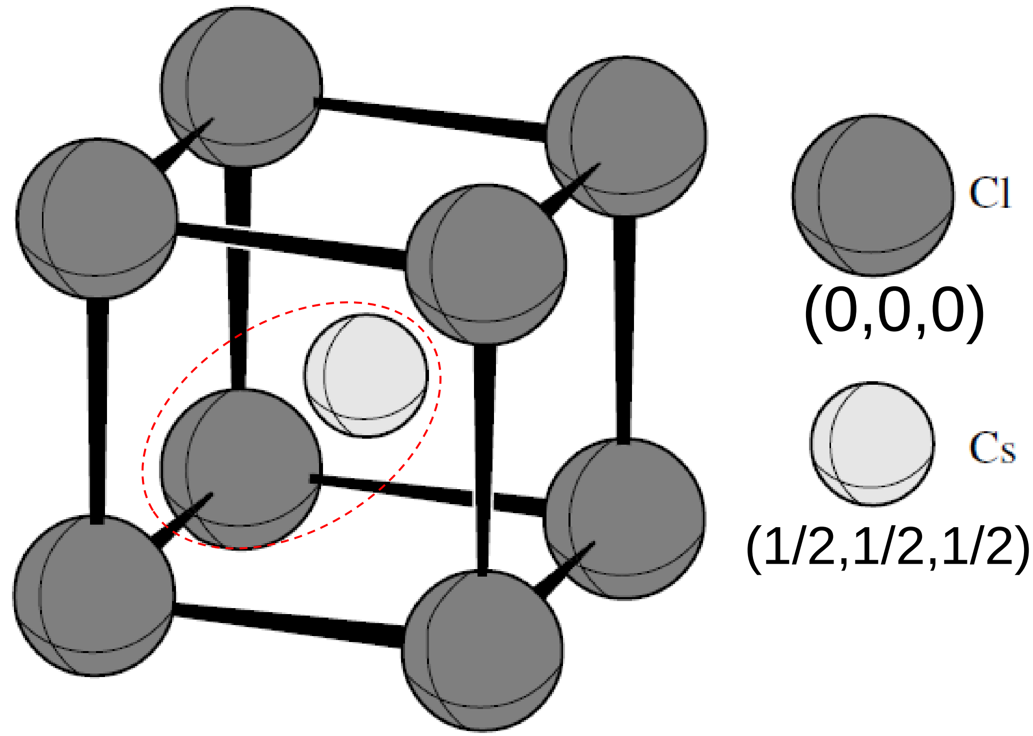
Formerly SC, FCC & BCC

These are the 3D Bravais lattices that all known crystals belong to!

Note: SC, BCC and FCC elemental metal structures are special cases of cubic P, I and F, respectively, where basis = one atom and each atom sits on a lattice point.

Bravais lattice + Basis in 3D



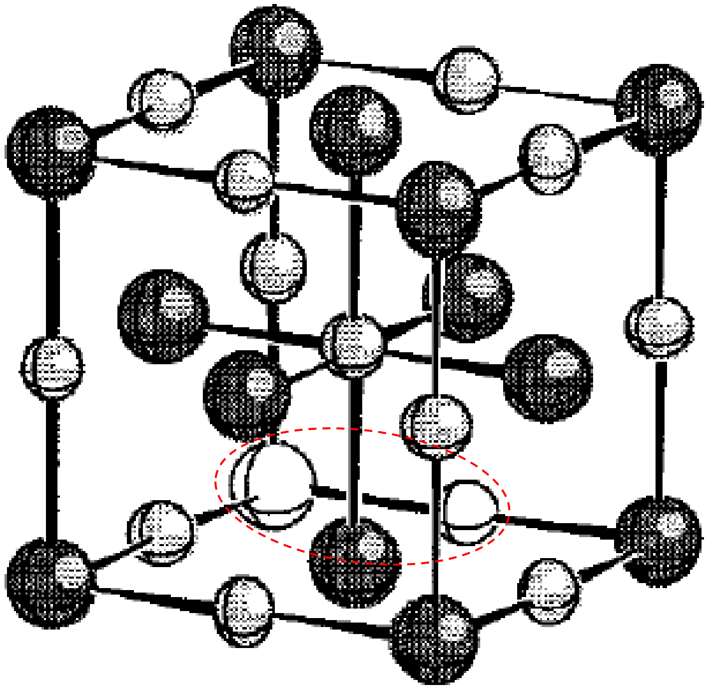


Cubic I BCC (elemental metal) **Cubic P** with diatom basis CsCl

2 lattice points per unit cell: Basis: (0,0,0) & (½, ½, ½)

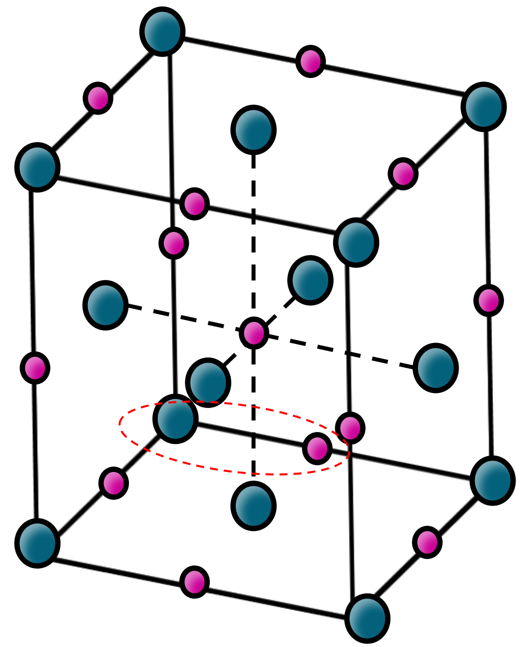
➔*All structures consist of one of the 14 Bravais lattices plus a Basis, i.e., a collection of atoms that decorate each lattice point. All lattice points must be identical! The atomic position does not need to coincide with a lattice point.*

Bravais lattice + Basis in 3D



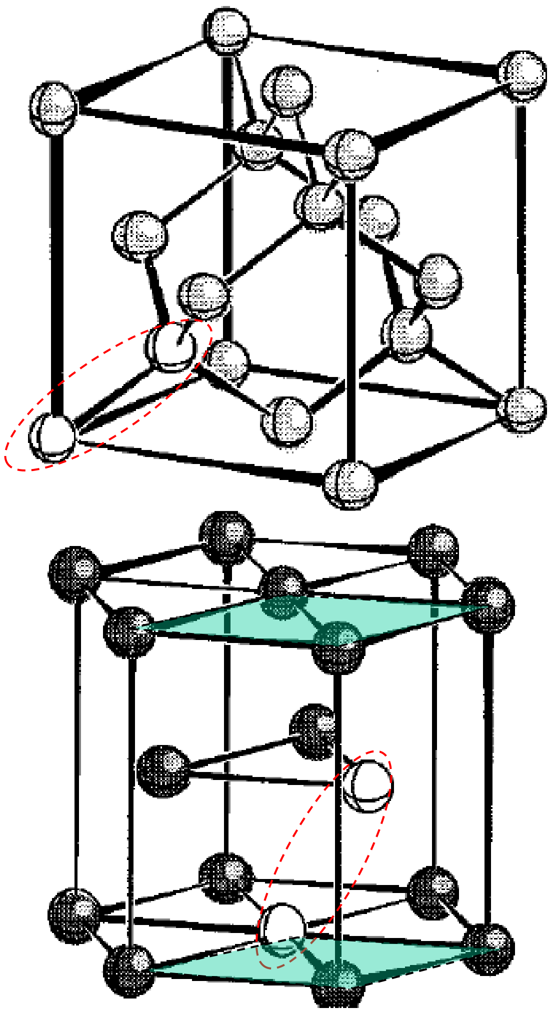
Note: Basis atoms in white

Rocksalt (NaCl):
Cubic F lattice with two atoms in basis
Related by a (1/2,0,0) displacement



<u>Cubic F centering + basis of 2 atoms:</u>
4 lattice points per unit cell × **2**
Lattice coordinates used for **Na^+**: (0,0,0); 8 corners per cube implicit; (½,½,0), (½,0,½), (0,½,½); 6 faces per cube implicit
Four additional coordinates used for **Cl^-**: one body center atom; 3 edge atoms; 12 edges per cube implicit

Bravais lattice + Basis in 3D



Diamond:
Cubic F lattice with two atoms in basis
Related by a displacement of (¼,¼,¼)
Additional atom is identical (e.g., C or Si) and occupies 4 tetrahedral sites: 8 atoms per unit cell in total

Hexagonal closed packed (HCP):
Hexagonal P lattice with two atoms per basis
Related by displacement (1/3, 2/3, 1/2)

Note: Basis atoms in white

3D Bravais lattices

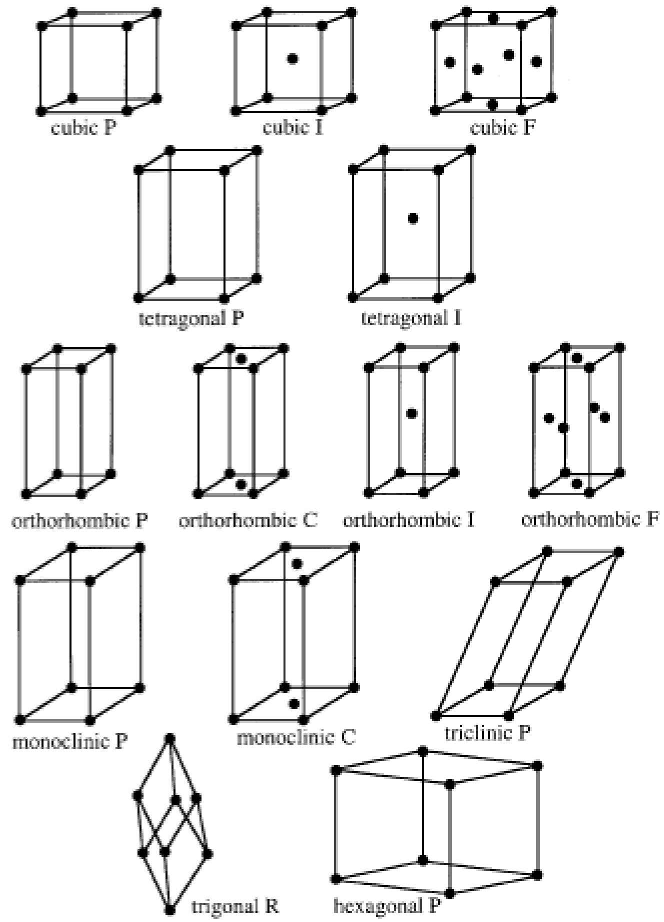
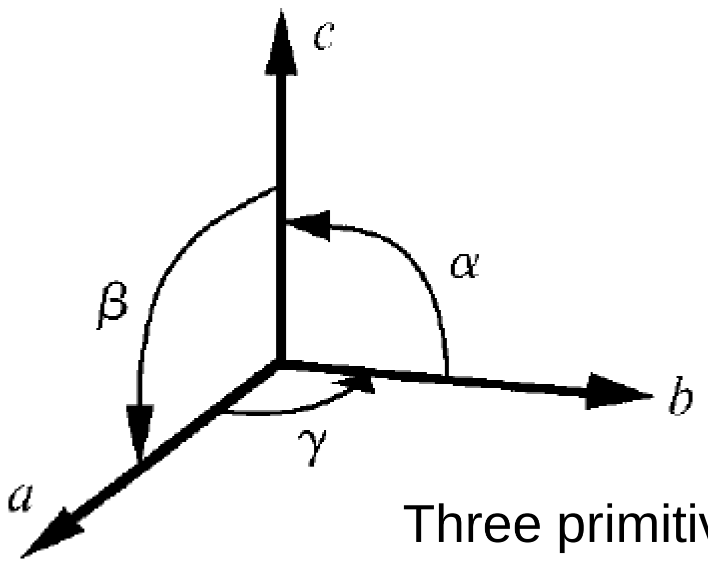


Figure 2.12. The 14 Bravais lattices.

Table 2.1. *Parameters for the Bravais lattices.*

System	Number of lattices	Lattice symbols	Restrictions on conventional cell axes and angles
Cubic	3	P, I, F	$a=b=c$ $\alpha=\beta=\gamma=90°$
Tetragonal	2	P, I	$a=b\neq c$ $\alpha=\beta=\gamma=90°$
Orthorhombic	4	P, C, I, F	$a\neq b\neq c$ $\alpha=\beta=\gamma=90°$
Monoclinic	2	P, C	$a\neq b\neq c$ $\alpha=\gamma=90°\neq\beta$
Triclinic	1	P	$a\neq b\neq c$ $\alpha\neq\beta\neq\gamma$
Trigonal	1	R	$a=b=c$ $\alpha=\beta=\gamma<120°,\neq 90°$
Hexagonal	1	P	$a=b\neq c$ $\alpha=\beta=90°,\ \gamma=120°$

Conventional lattice parameter definition



$$\mathbf{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$$

Three primitive vectors, ***a*, *b*, *c***
Six scalars:
- Magnitude of primitive vectors (a, b, c)
- Relative orientation of primitive vectors (α,β,γ)

Note: reference frame is Cartesian only for orthonormal systems

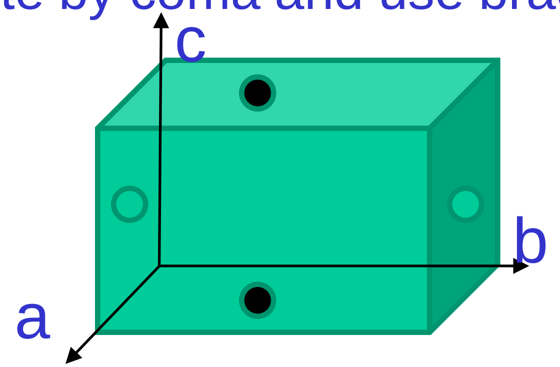
Location in a unit cell

$\mathbf{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$ ➔Definition of the Bravais lattice

$\mathbf{r}_j = x_j\mathbf{a} + y_j\mathbf{b} + z_j\mathbf{c}$ ➔Position of the j atoms within the unit cell with respect to a Bravais lattice origin

(x_j, y_j, z_j)

➔Positions within the unit cell are identified by coefficients of location
➔Three fractional coordinates (*x*, *y*, *z*)
➔Greater than or equal to zero, but less than 1: $0 \leq (x, y, z) < 1$
➔No unit needed
➔Separate by coma and use brackets (parantheses)



$(0.5, 0, 0.5)$

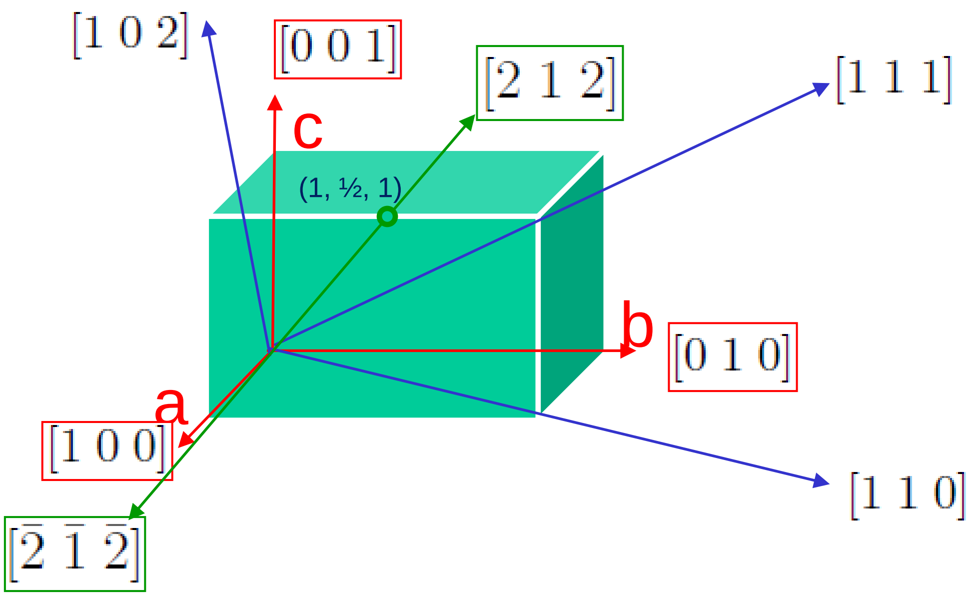
$(0.5, 0.3, 0)$

Notice that (0.5,0,0.5) and (0.5,1,0.5) are equivalent.
(0.5,0.3,0) and (0.5,0.3,1) are also equivalent.

Unit cell: implies periodic boundary conditions on all sides

Directions in a unit cell

➔Three integer indices $[u\ v\ w]$
➔Component of the Bravais lattice that specifies the direction of interest



Algorithm

1. Read off point coordinates of ***intercept*** with unit cell and ***tail*** in terms of unit cell dimensions *a*, *b*, and *c*
2. Calculate difference (**Intercept** – **Tail**)
3. Adjust to smallest integer values
4. Enclose in square brackets, no commas [*uvw*]
5. Use overbar symbol to designate negative directions

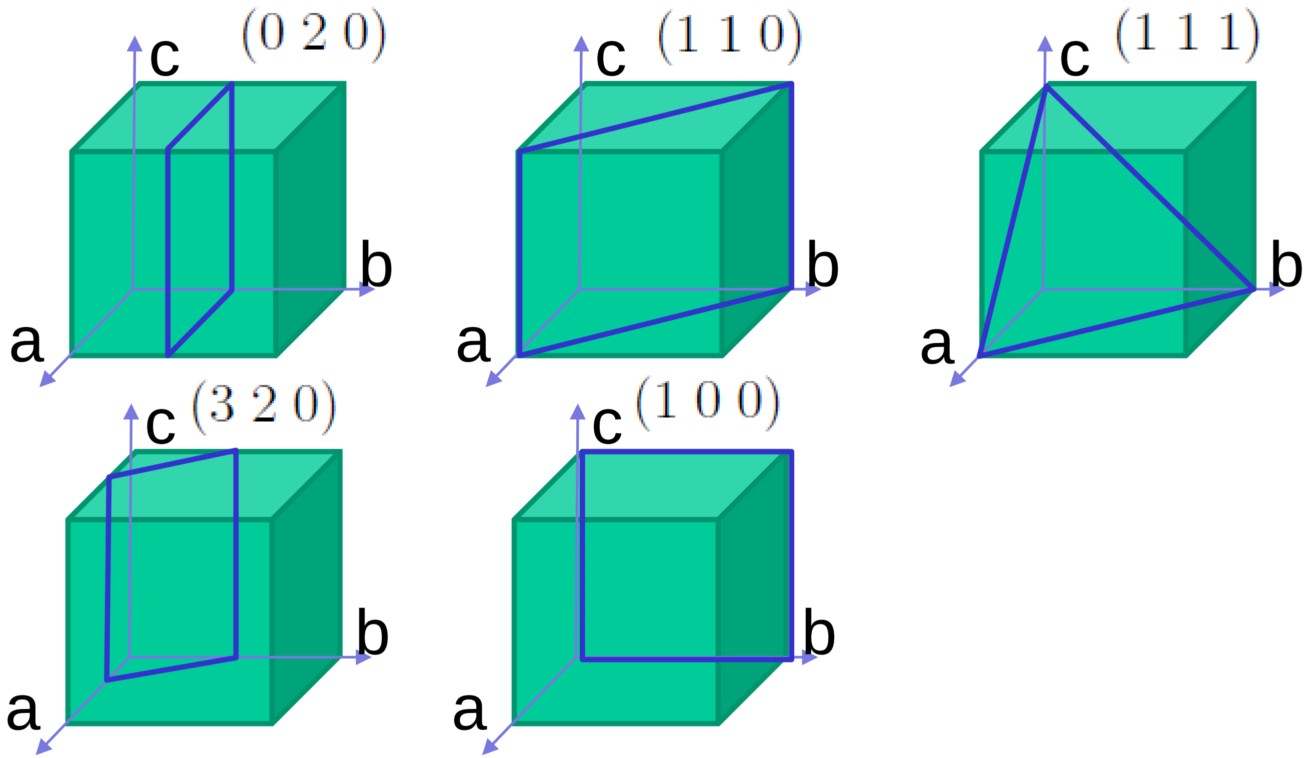
$[2\ 1\ 2]$ and $[\bar{2}\ \bar{1}\ \bar{2}]$ belong to the same family of directions, noted $\langle 2\ 1\ 2 \rangle$

The number of equivalent directions in a family depends on the Bravais lattice

Planes in a unit cell

➔Indices for planes
➔The reciprocal of the intercepts

$$(1/u\ 1/v\ 1/w)$$



In cubic crystals, $\begin{matrix}(1\ 0\ 0) & (\bar{1}\ 0\ 0) \\ (0\ 1\ 0) & (0\ \bar{1}\ 0) \\ (0\ 0\ 1) & (0\ 0\ \bar{1})\end{matrix}$ all belong to $\{1\ 0\ 0\}$

Crystallographic Planes

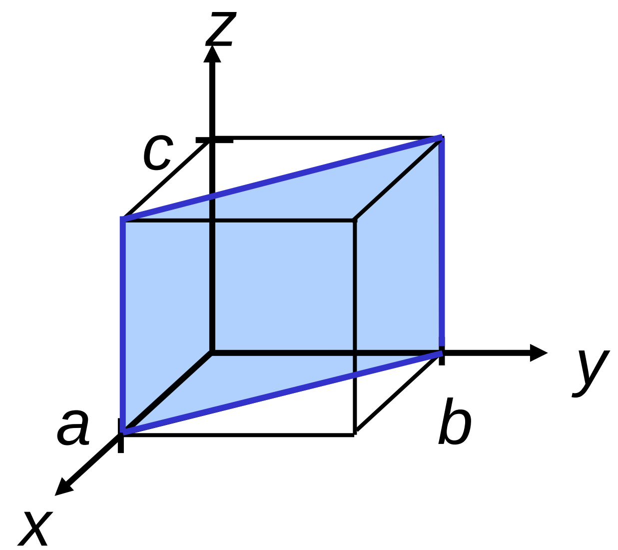
- Miller Indices: Reciprocals of the (three) axial intercepts for a plane, cleared of fractions & common multiples.

Algorithm

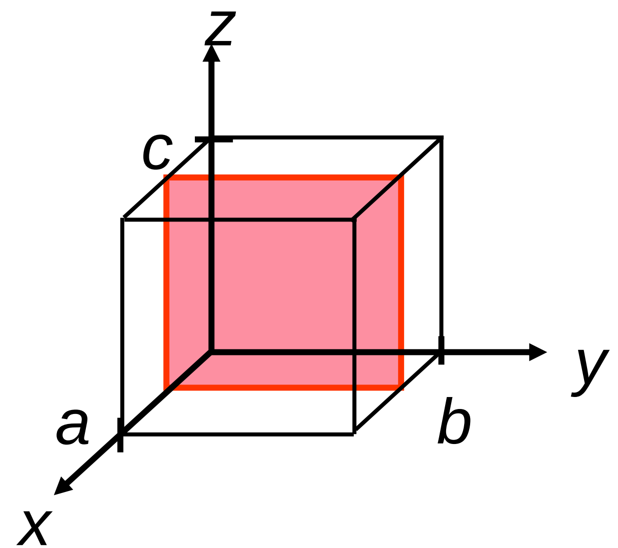
1. Planes cannot pass through the origin
2. Planes must intersect or be parallel to the axes
3. Read off intercepts of plane with axes in terms of *a*, *b*, *c*
4. Take reciprocals of intercepts
5. Reduce to smallest integer values
6. Enclose in parentheses, no commas, i.e., (*hkl*)

In-class exercise 1

<u>example</u>		*a*	*b*	*c*
1.	Intercepts	1	1	∞
2.	Reciprocals	1/1	1/1	1/∞
		1	1	0
3.	Reduction	1	1	0
4.	Miller Indices	(110)		

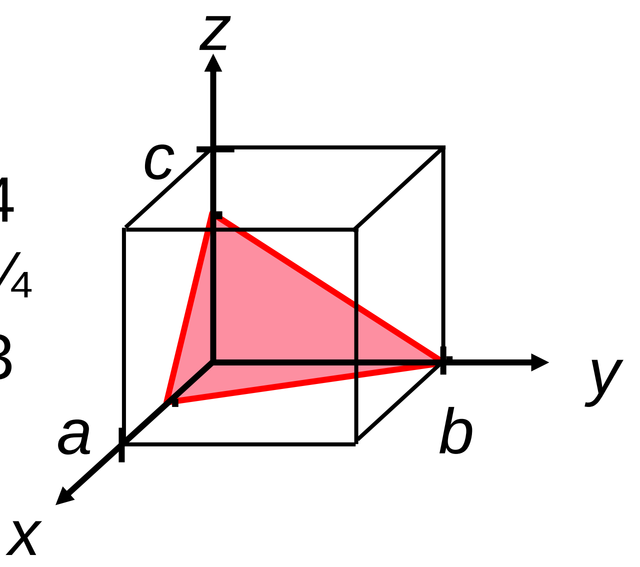


<u>example</u>		*a*	*b*	*c*
1.	Intercepts	1/2	∞	∞
2.	Reciprocals	1/½	1/∞	1/∞
		2	0	0
3.	Reduction	2	0	0
4.	Miller Indices	(200)		



In-class exercise 1

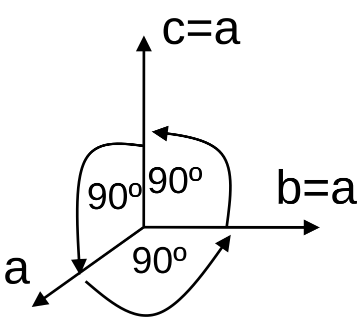
<u>example</u>	*a*	*b*	*c*
1. Intercepts	1/2	1	3/4
2. Reciprocals	1/½	1/1	1/¾
	2	1	4/3
3. Reduction	6	3	4
4. Miller Indices	(634)		



Cubic P Lattice

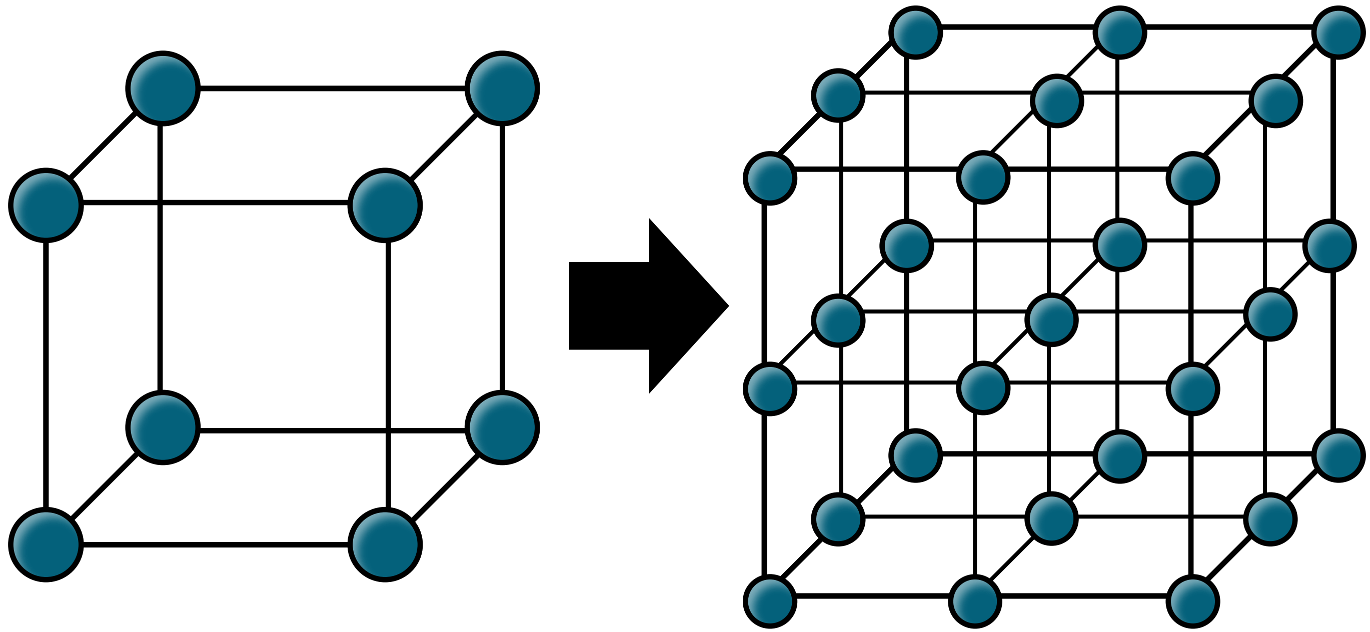
$$\mathrm{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$$

Any unit cell that encloses a volume V = **a**·(**b**×**c**) and has only one lattice point is designated as **Primitive (P)**



$$a = b = c$$

$$\alpha = \beta = \gamma = 90^o$$



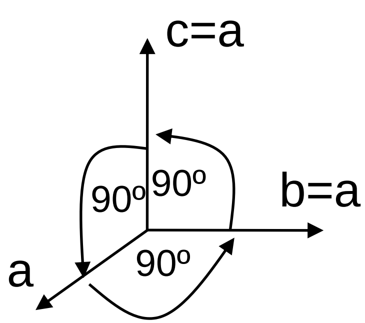
Cubic Primitive centering:

1 lattice point per unit cell

6 nearest neighbors (coordination number = 6)

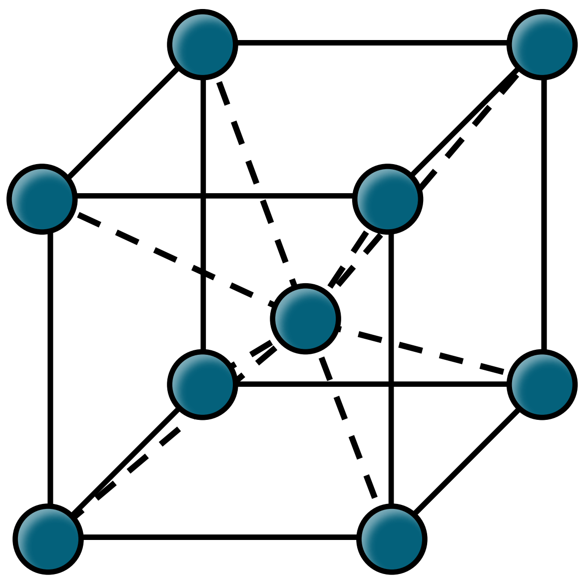
Single lattice coordinate: (0,0,0); 8 corners of cube implicit

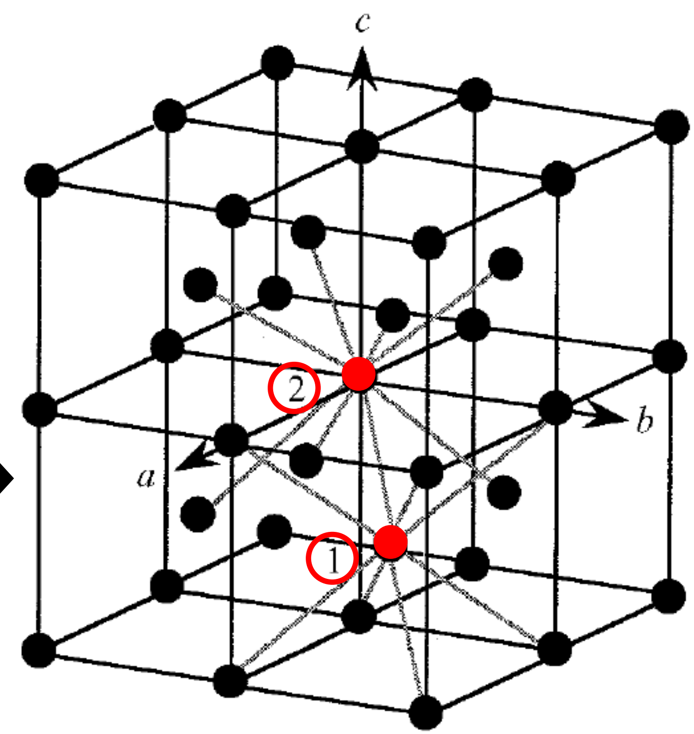
Cubic I Lattice



$a = b = c$

$\alpha = \beta = \gamma = 90^o$





1 and 2 have identical 8-fold coordination → Bravais lattice

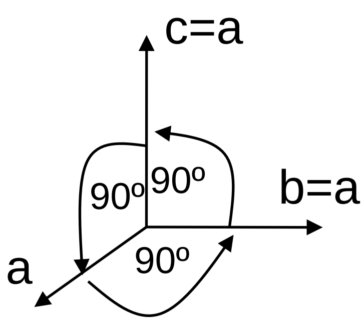
<u>Cubic I centering:</u>

2 lattice points per unit cell

8 nearest neighbors (coordination number = 8)

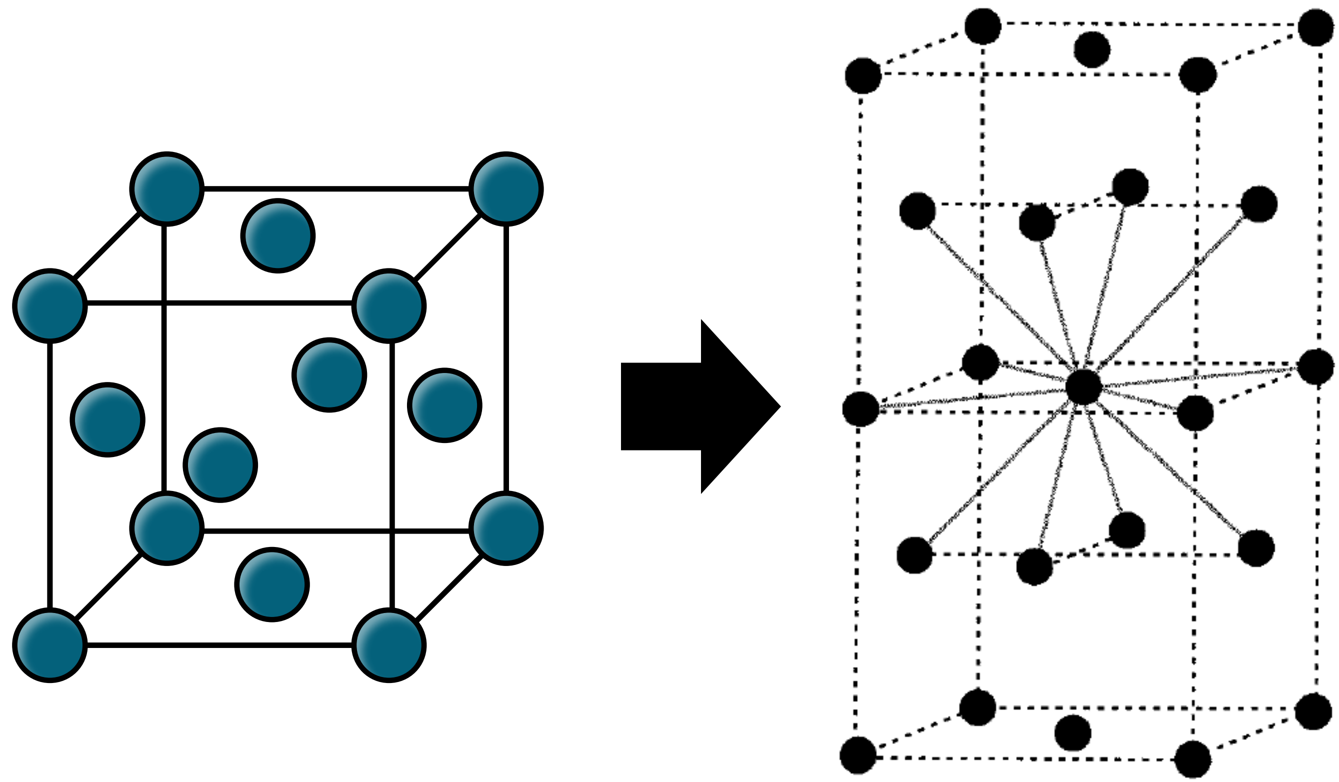
Two lattice coordinates: (0,0,0); 8 corners per cube implicit; (½,½,½) body center of cube implicit

Cubic F Lattice



$$a = b = c$$

$$\alpha = \beta = \gamma = 90^o$$



Cubic F centering:

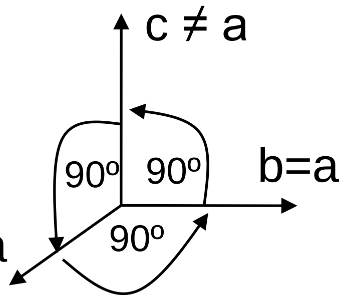
4 lattice points per unit cell

12 nearest neighbors (coordination number = 12)

Lattice coordinates: (0,0,0); 8 corners per cube implicit;

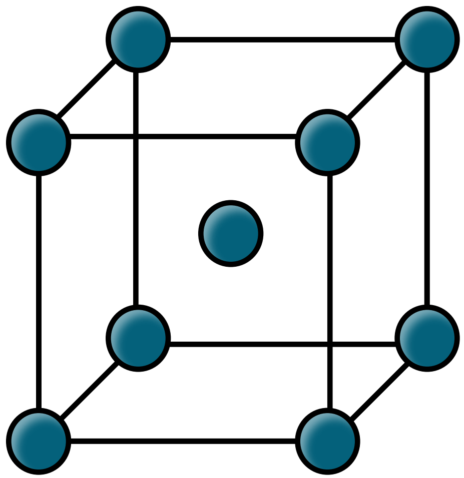
(½,½,0), (½,0,½), (0,½,½); 6 faces per cube implicit

Tetragonal P / I Lattices



$$a = b \neq c$$

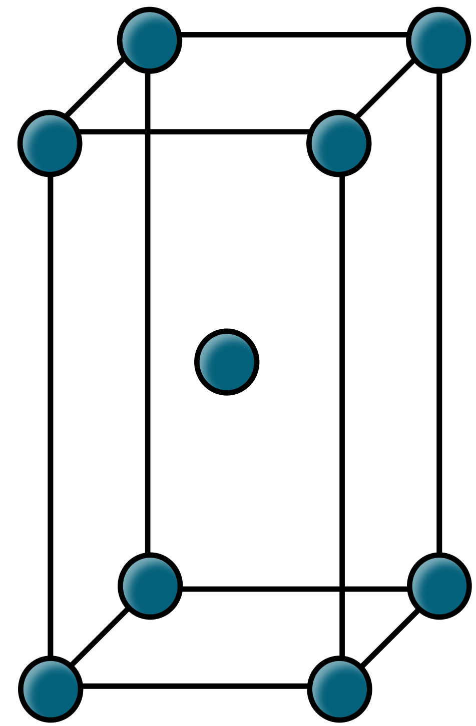
$$\alpha = \beta = \gamma = 90^o$$



Cubic P

Cubic I

Stretching along [001] axis

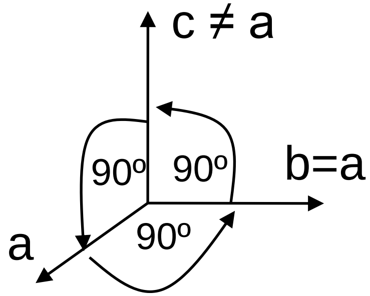


Tetragonal P

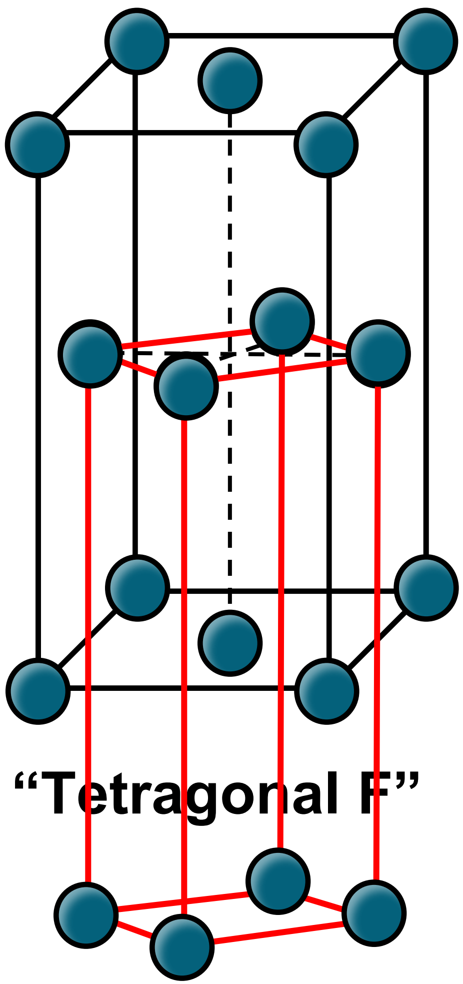
Tetragonal I

Tetragonal F Lattice?

Stretching along [001] axis the Cubic F?

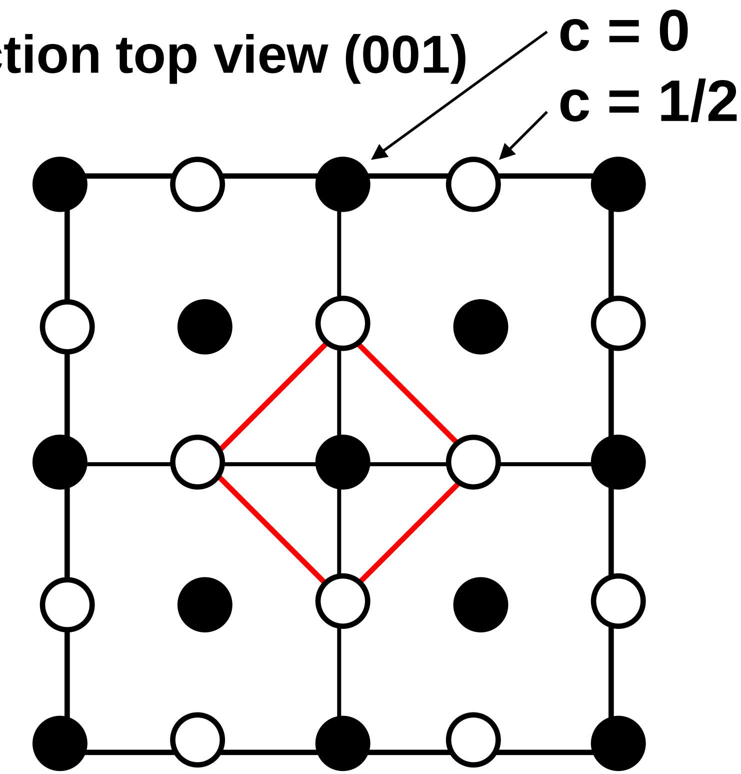


$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^o$$



"Tetragonal F"

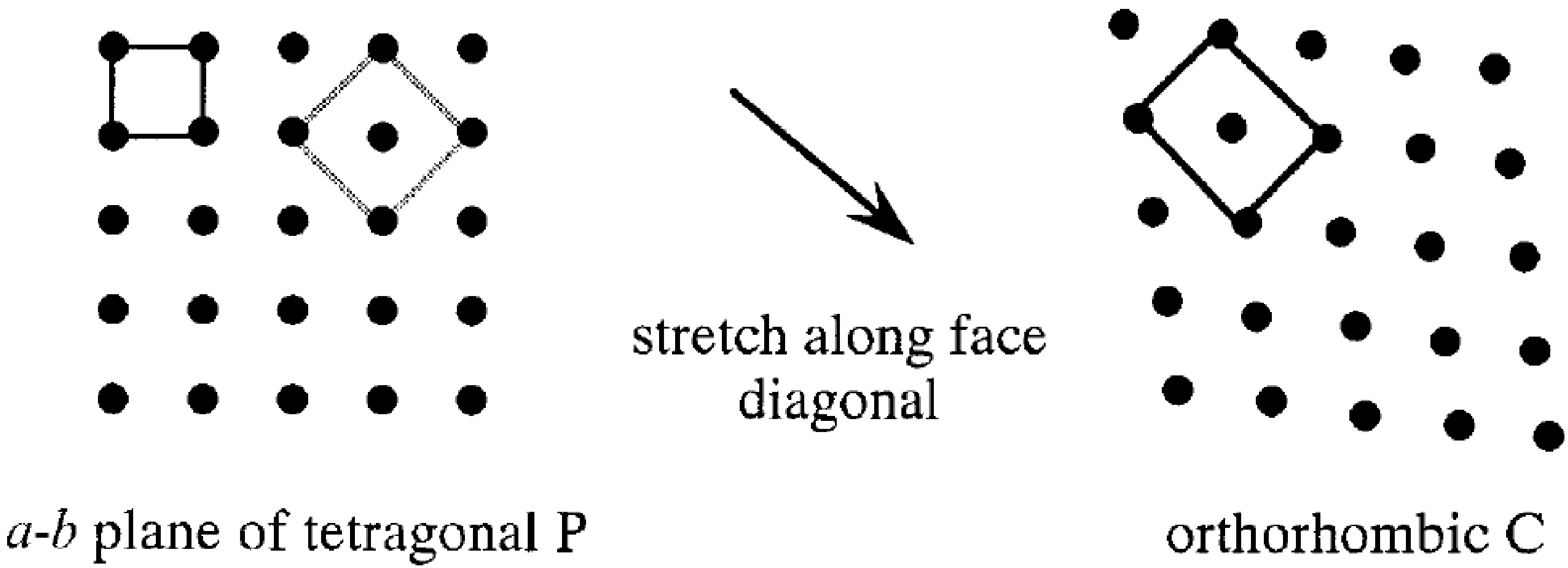
Projection top view (001)



"Tetragonal F"→Tetragonal I

Tetragonal to orthorhombic

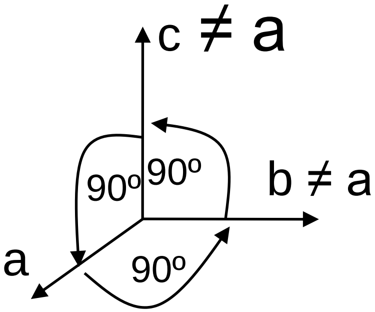
Let's lower the symmetries a bit more

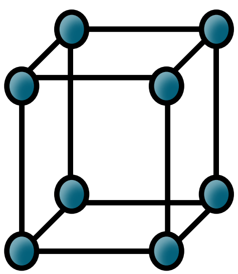


Projection top view (001)

Orthorhombic

Stretch along [010] & [001] by different amounts

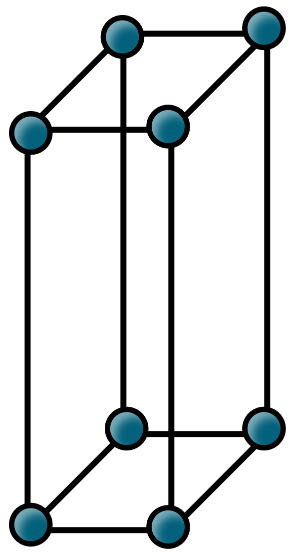




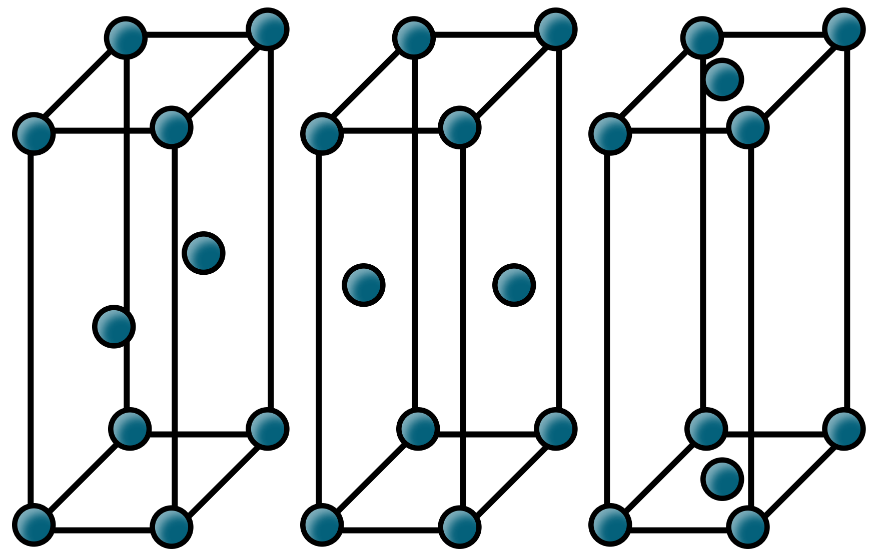
Cubic P

$a \neq b \neq c$

$\alpha = \beta = \gamma = 90^o$



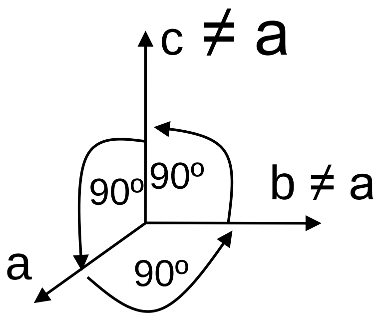
Orthorhombic P



Orthorhombic A, B and C

# lattice pts	**1**	**2**	**2**	**2**
	(0,0,0)	(0,0,0)	(0,0,0)	(0,0,0)
		(0,1/2,1/2)	(1/2,0,1/2)	(1/2,1/2,0)
		c-b plane	a-c plane	a-b plane

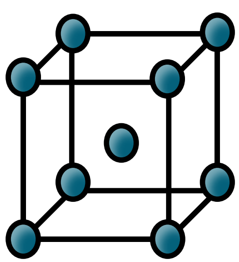
Orthorhombic



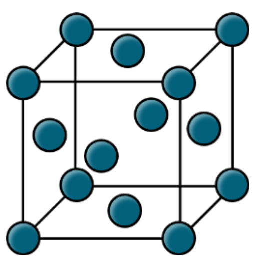
Stretch along [010] & [001] by different amounts

$$a \neq b \neq c$$

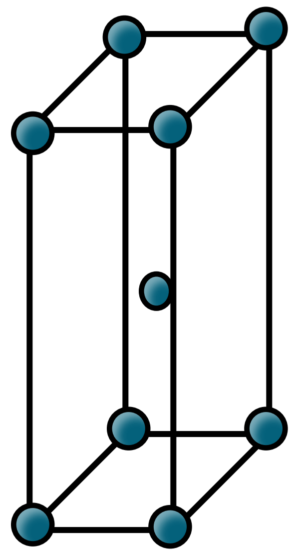
$$\alpha = \beta = \gamma = 90^o$$



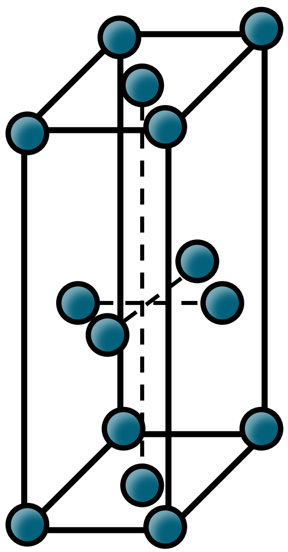
Cubic I



Cubic F



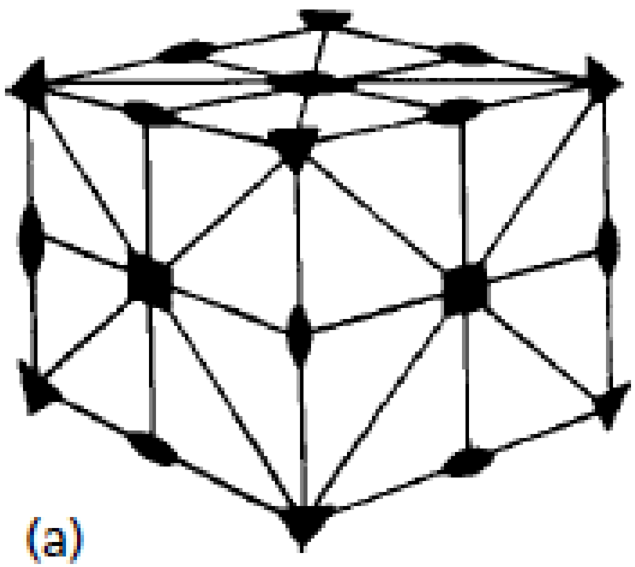
Orthorhombic I

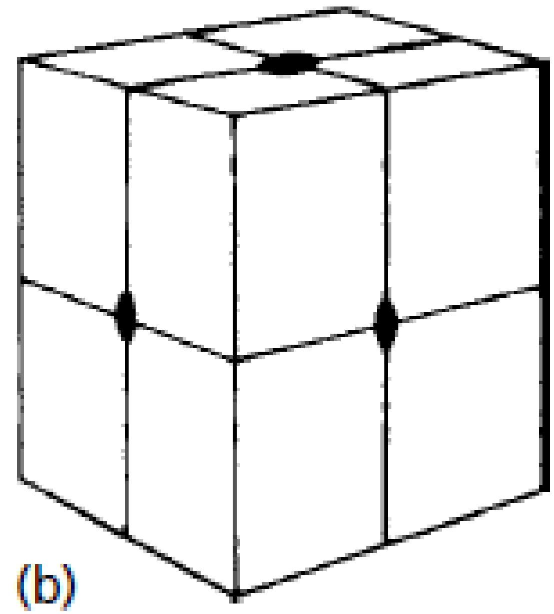


Orthorhombic F

# lattice pts	Orthorhombic I	Orthorhombic F
	2	**4**
	(0,0,0)	(0,0,0)
	(1/2,1/2,1/2)	(1/2,1/2,0) (1/2, 0,1/2) (0, 1/2, 1/2)

Crystal systems & symmetries





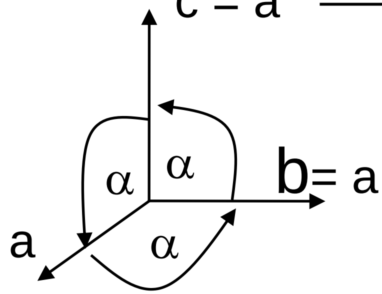
cubic

orthorhombic

Lower symmetry than cubic

More on 3D symmetries and diffraction in future topics!

Trigonal lattice



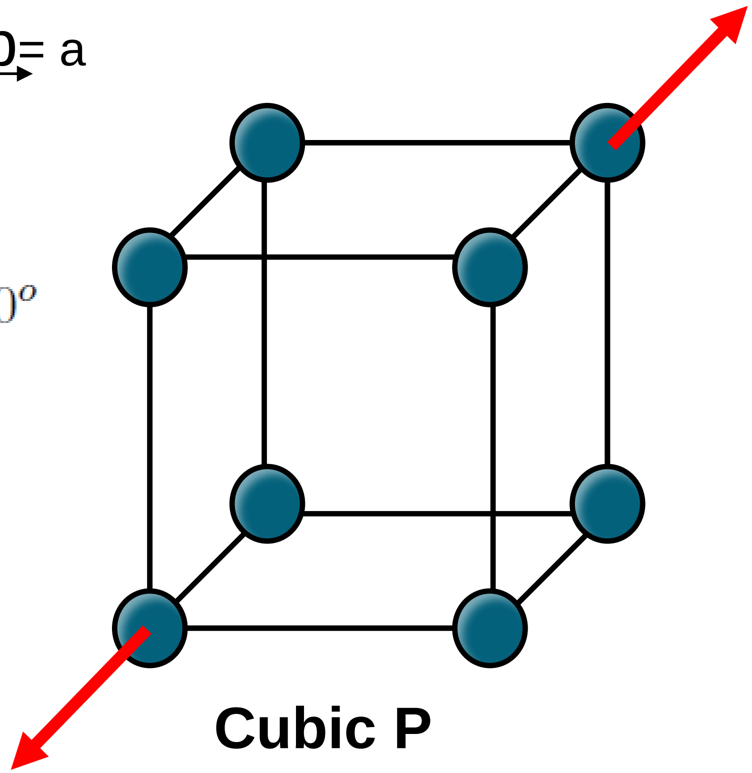
$a = b = c$

$\alpha = \beta = \gamma \neq 90^o$

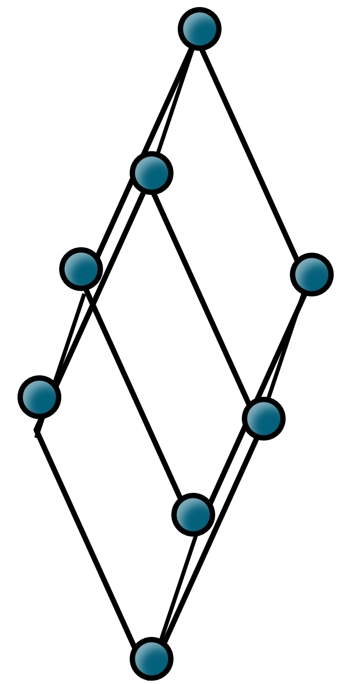
< 120°

Distortion of the angles, not lattice dimensions

Stretch along [111]

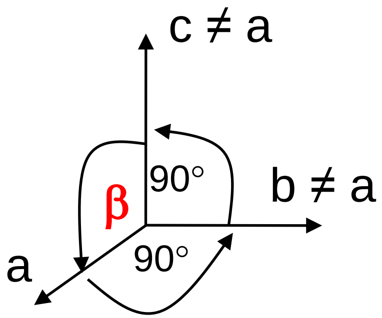


Cubic P



Trigonal R (rhombohedral)

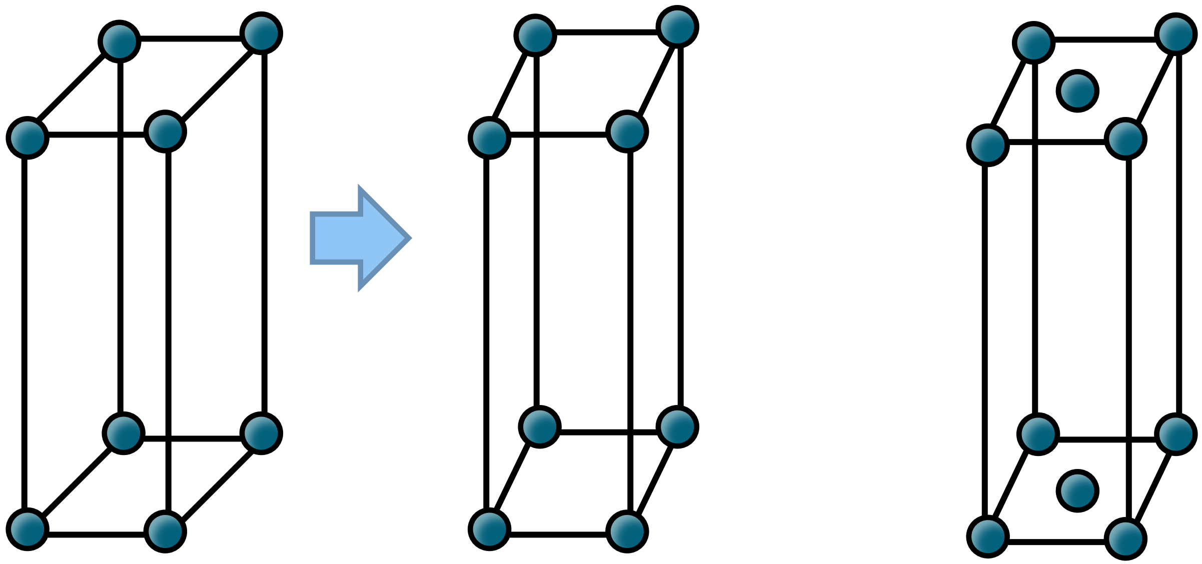
Monoclinic crystals



$\alpha = \gamma = 90^o, \; \beta \neq 90^o$

(modern convention)

Distortion of the angles



Orthorhombic P **Monoclinic P** **Monoclinic C=A**

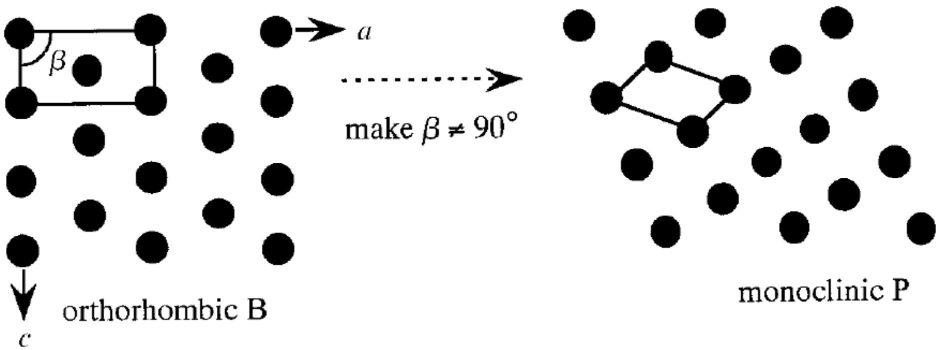
$a \neq b \neq c$

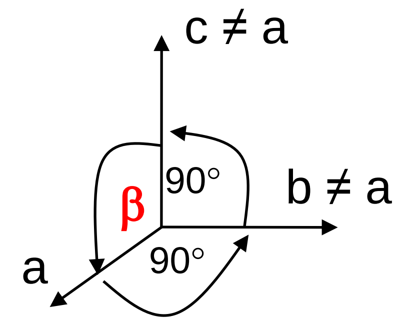
$\alpha = \beta = \gamma = 90^o$

$a \neq b \neq c$

$\alpha = \gamma = 90^o, \; \beta \neq 90^o$

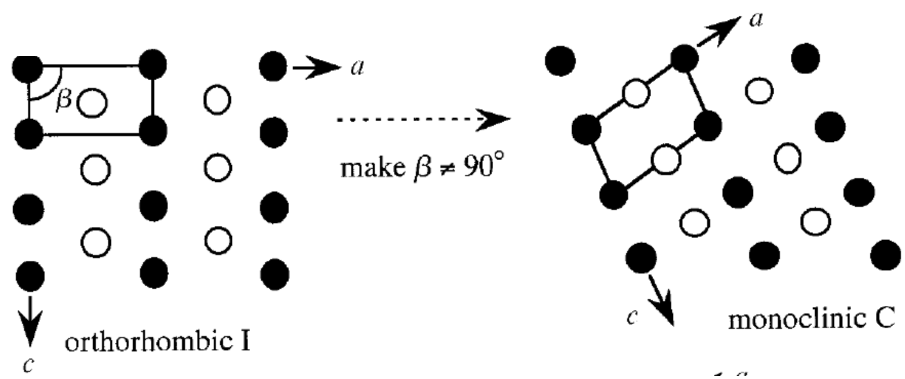
Orthorhombic to monoclinic

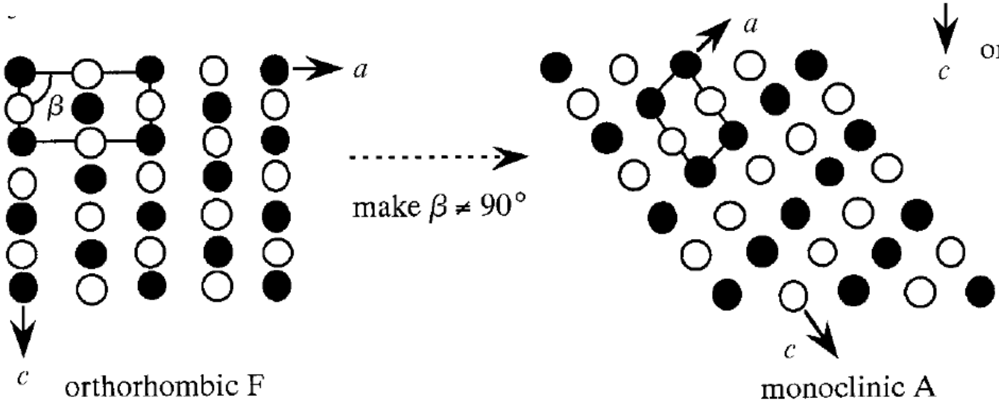




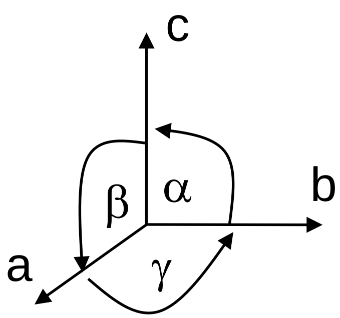
$$\alpha = \gamma = 90^o, \ \beta \neq 90^o$$

Projection view along [010] (b-axis) at (101) plane



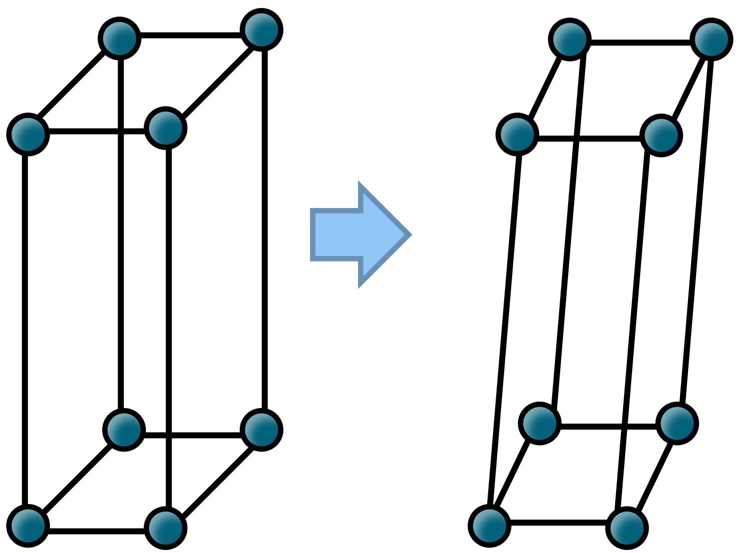


Triclinic crystals



$$\alpha \neq \gamma \neq \beta \neq 90^o$$

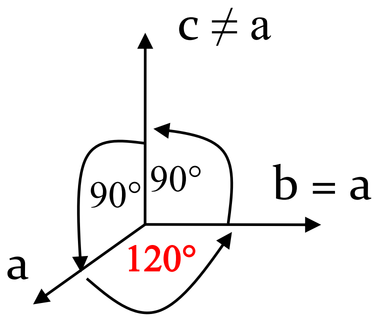
<u>Distortion of all angles</u>



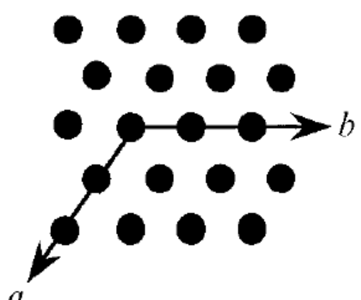
Orthorhombic P **Triclinic P**

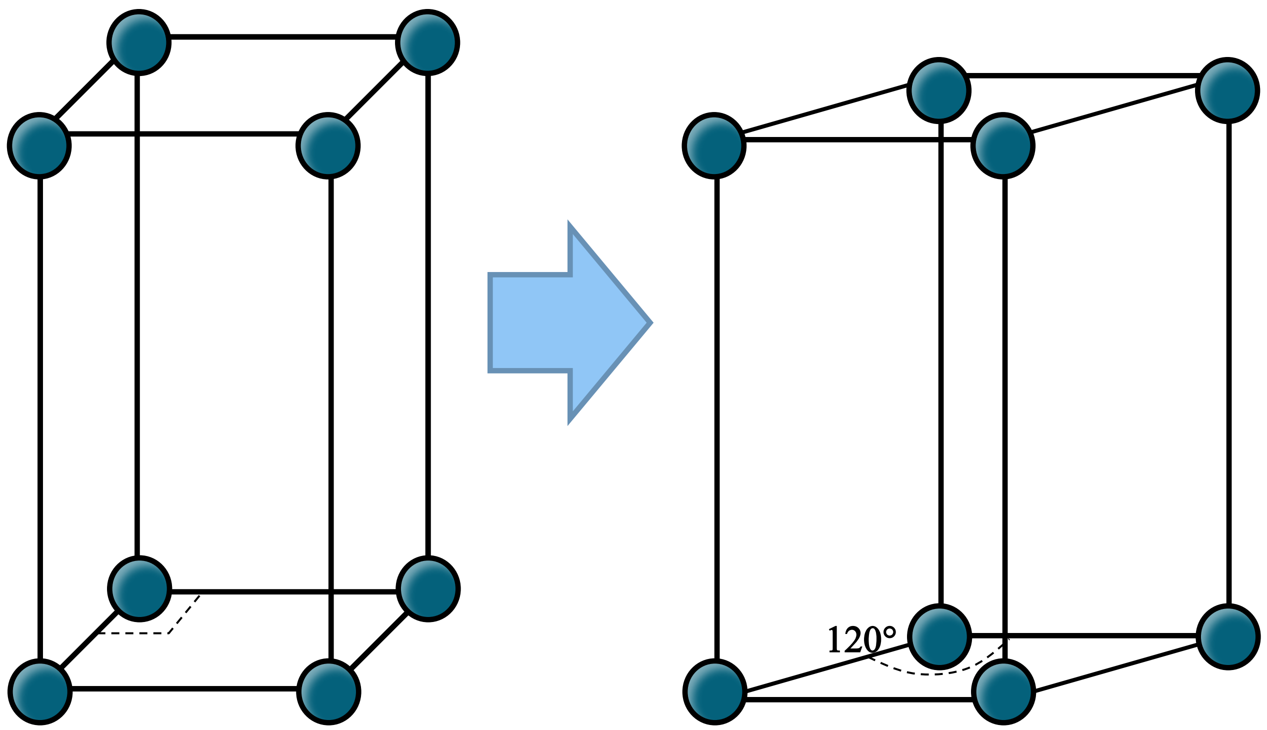
Among 26000 inorganic compounds
➔27.6% are Cubic, 72% are Orthorhombic or higher symmetry
Only a minority are Triclinic ➔ Nature prefers symmetry

Hexagonal lattice



$$a = b \neq c$$
$$\alpha = \beta = 90^o, \ \gamma = 120^o$$





Tetragonal P **Hexagonal P**

Planes in Hexagonal Crystals

Hexagonal crystals → Miller-Bravais indices

Directions and planes in hexagonal lattices and crystals can be designated by the *4-index Miller-Bravais notation* instead of 3-index Miller indices

In the four-index notation:

- the first three indices are a symmetrically related set on the basal plane
- the third index is a *redundant one* (which can be derived from the first two) and is introduced to make sure that members of a family of directions or planes have a set of numbers which are identical - this is because in 2D two indices *suffice* to describe a lattice (or crystal)
- the fourth index represents the 'c' axis *(⊥ to the basal plane)*

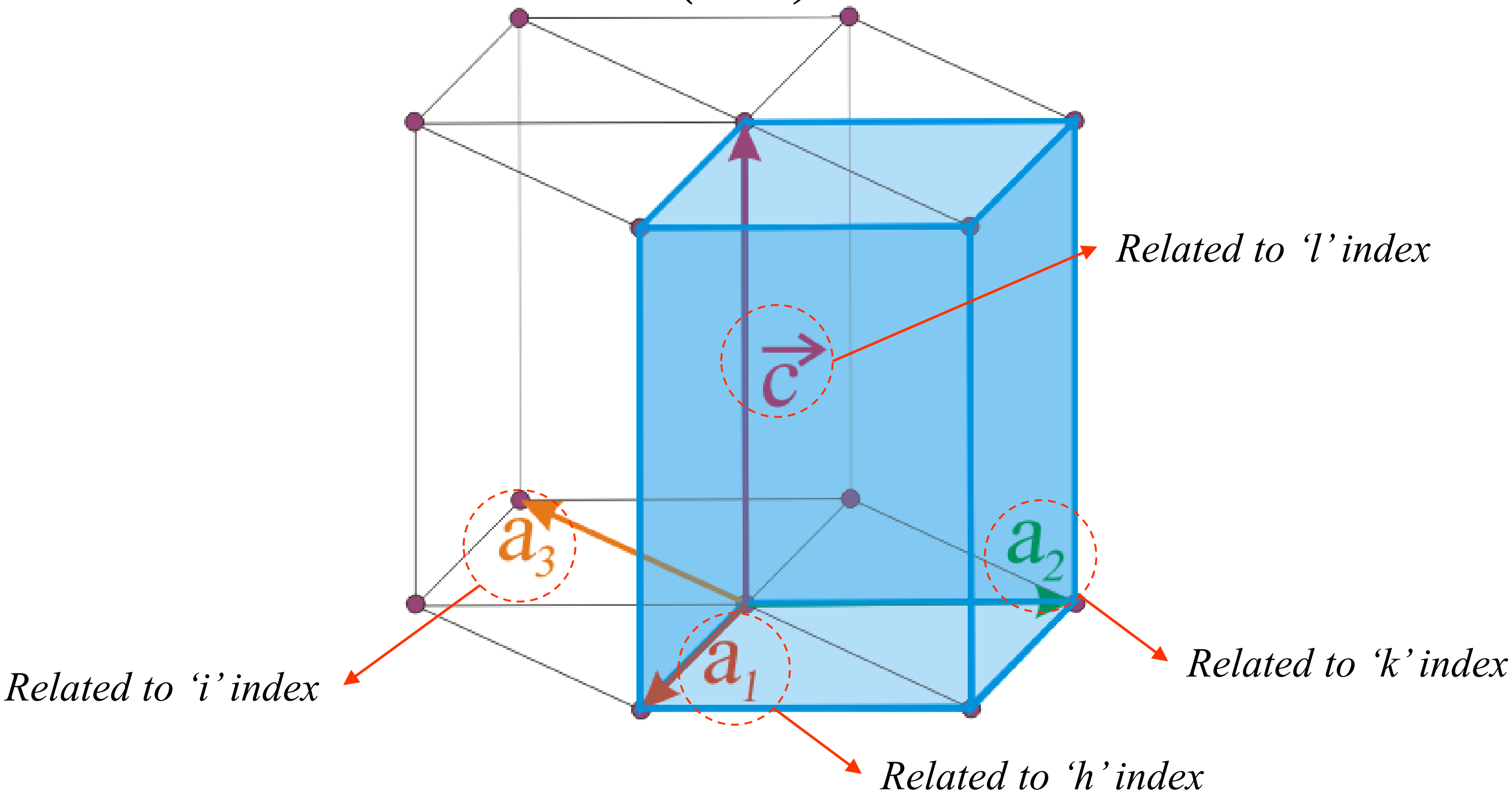
The first three indices in a hexagonal lattice can be permuted to get the different members of a family; while, the fourth index is kept separate.

Planes in Hexagonal Crystals

Miller-Bravais notation

$(h\ k\ i\ l)$

$i = -(h + k)$

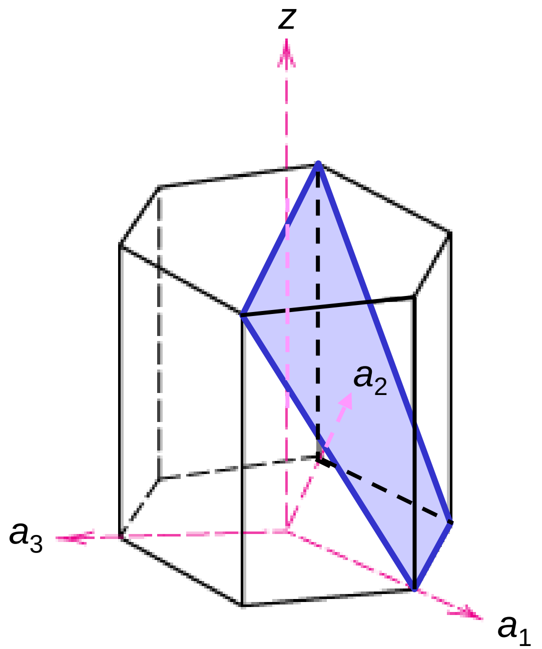


In class exercise 2

- In hexagonal unit cells the same idea is used

<u>example</u>		a_1	a_2	a_3	c
1.	Intercepts	1	∞	-1	1
2.	Reciprocals	1	1/∞	-1	1
		1	0	-1	1
3.	Reduction	1	0	-1	1

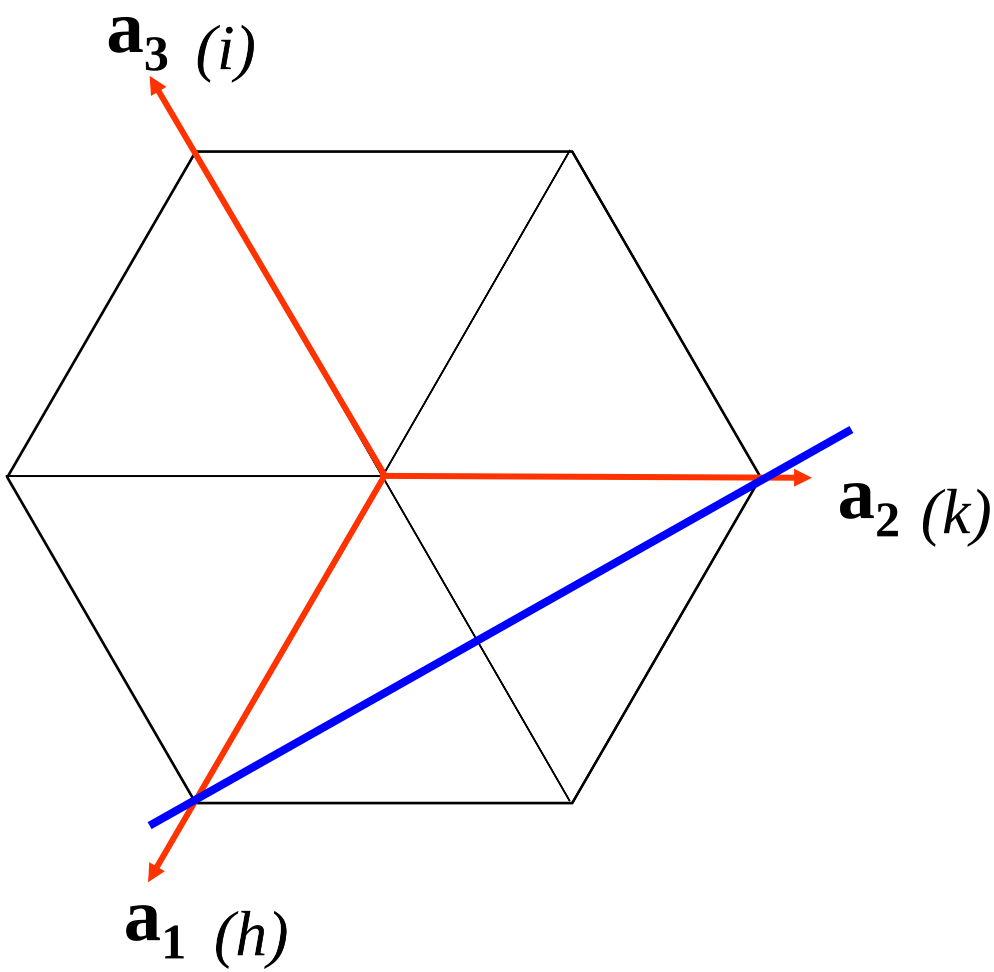
4. Miller-Bravais Indices $(10\bar{1}1)$



Adapted from Fig. 3.8(a), *Callister 7e.*

$$(h\ k\ i\ l)$$
$$i = -(h + k)$$

In class exercise 3



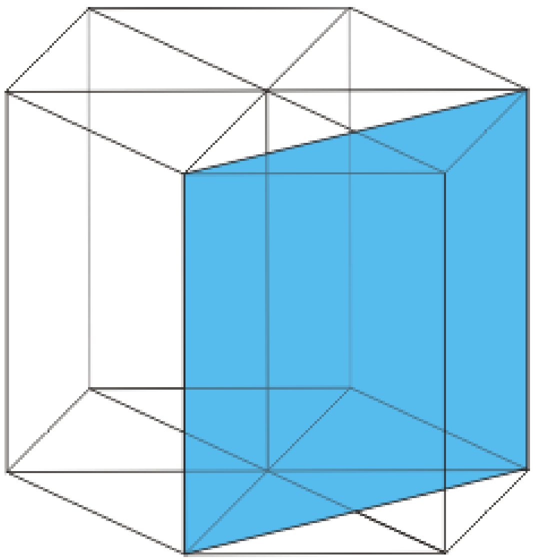
$(h\ k\ i\ l)$, $i = -(h + k)$

Intercepts $\rightarrow$ $1\ 1\ -\tfrac{1}{2}\ \infty$

Plane $\rightarrow$ $(1\ 1\ \bar{2}\ 0)$

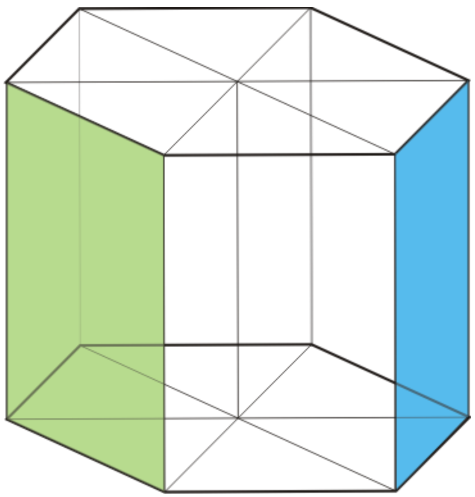
$(h\ k\ i\ l)$

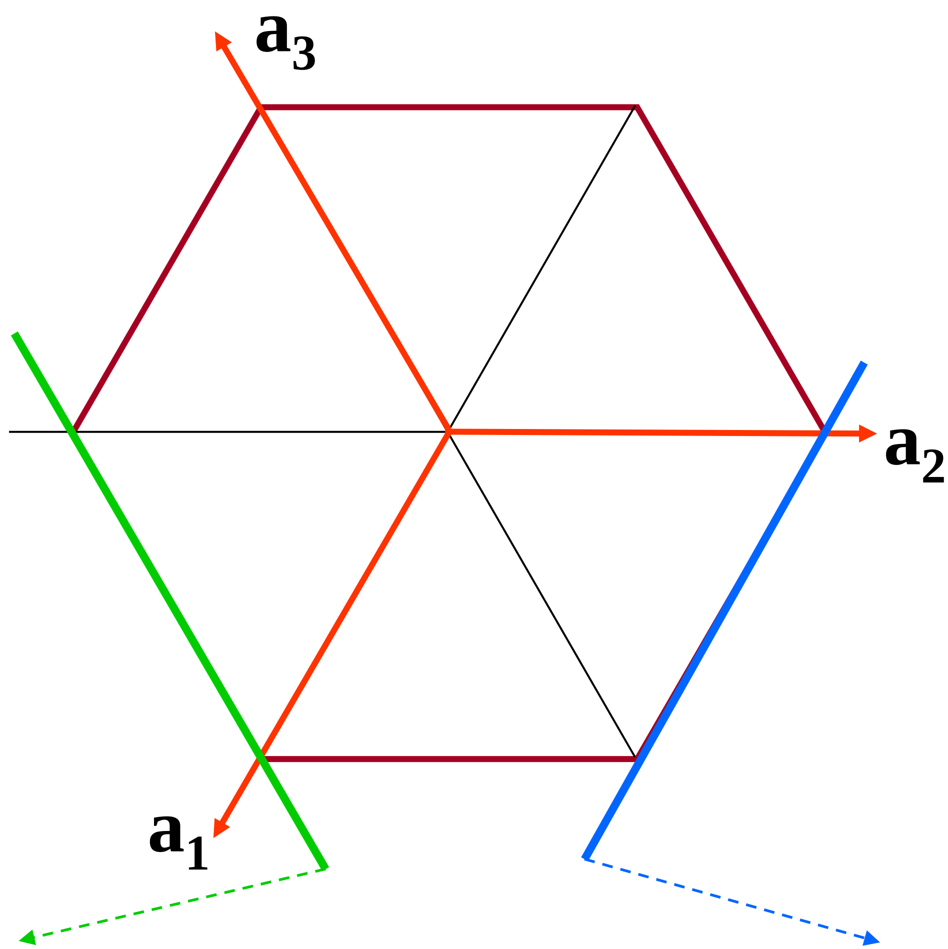
$i = -(h + k)$



The use of the 4-index notation is to bring out the equivalence between **crystallographically equivalent planes and directions**

In class exercise 4





Intercepts → $1\ {-1}\ \infty\ \infty$
Miller → $(1\ \bar{1}\ 0)$
Miller-Bravais → $(1\ \bar{1}\ 0\ 0)$

Intercepts → $\infty\ 1\ {-1}\ \infty$
Miller → $(0\ 1\ 0)$
Miller-Bravais → $(0\ 1\ \bar{1}\ 0)$

$(h\ k\ i\ l)$
$i = -(h + k)$

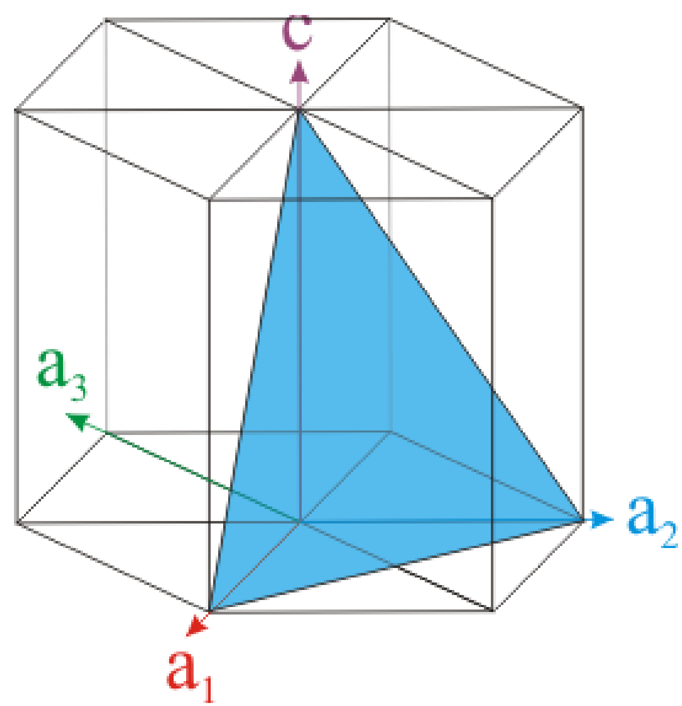
In class exercise 5

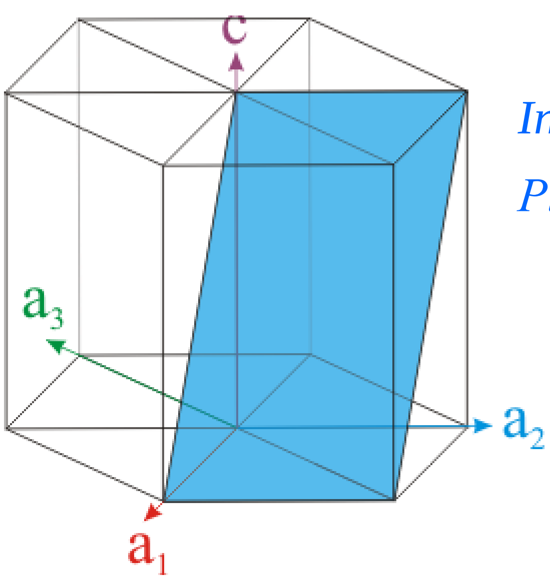
$(h\ k\ \textcolor{red}{i}\ l)$

$i = -(h + k)$

Intercepts $\rightarrow 1\ 1\ \text{-}\ ½\ 1$

Plane $\rightarrow (1\ 1\ \bar{2}\ 1)$





Intercepts $\rightarrow 1\ \infty\ -1\ 1$

Plane $\rightarrow (1\ 0\ \bar{1}\ 1)$